Arista Access Node (YEL-ACC-32) Hardware/Environmental Fault Resolution SOP

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7159

Description

The Arista Access Node (YEL-ACC-32) at Yelahanka Sub-station has experienced a hardware/environmental fault, manifesting as ASIC memory parity errors, line card crashes, and power supply redundancy loss, as indicated by Alarm Code ALARM_7159 with a Minor severity. This failure domain occurs due to the inherent limitations of the Arista access node's design, which relies on the EOS (Extensible Operating System) to manage and control the ASIC (Application-Specific Integrated Circuit). In the event of an ASIC memory parity error or line card crash, the EOS may attempt to recover the node, but if the power supply redundancy is lost, the node may become unresponsive. The underlying protocol mechanics, specifically the use of EOS's distributed architecture, can exacerbate the issue by propagating errors throughout the control plane, leading to a cascade of failures.

Pre-requisite Checks

- Verify the site location and Node ID (YEL-ACC-32) to confirm the affected node.

- Check the redundant link status to ensure it is operational and not contributing to the fault.

- Dispatch remote hands to Yelahanka Sub-station to visually inspect the node and environment for any signs of physical damage or excessive heat.

Potential Root Causes

Cause	Probability
ASIC memory parity error or line card crash	High
Power supply redundancy loss	Medium
EOS software bug or misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource utilization, Expected Output: System resource utilization metrics, including CPU, memory, and interface usage., Description: This command is used to isolate the root cause of the fault by analyzing system resource utilization metrics. Elevated CPU, memory, or interface usage can indicate an ASIC memory parity error or line card crash.

Command: show system alarm history, Expected Output: Alarm history, including alarm codes, timestamps, and severity levels., Description: This command is used to verify the occurrence of the ALARM_7159 code and understand the sequence of events leading to the fault.
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```
Command: show hardware alarm, Expected Output: Hardware alarm details, including alarm codes, timestamps, and severity levels., Description: This command is used to isolate the root cause of the fault by analyzing hardware alarm details. The presence of a hardware alarm can indicate a line card crash or power supply redundancy loss.
```

Resolution Steps

Step 1: Step: Verify Node ID and Site Location, Command: show system node, Description: Verify the Node ID (YEL-ACC-32) and site location to confirm the affected node.

Step 2: Step: Replace faulty ASIC or line card, Command: hw-module replace, Description: Replace the faulty ASIC or line card with a new one to resolve the fault. This step requires physical access to the node and may require remote hands assistance.

Step 3: Step: Restore power supply redundancy, Command: power supply redundancy on, Description: Restore power supply redundancy to prevent future faults. This step requires physical access to the node and may require remote hands assistance.

Post-Check Verification

```
Command: show system resource utilization, Expected Output: System resource utilization metrics, including CPU, memory, and interface usage., Description: Verify that system resource utilization metrics are within normal ranges to ensure the fault has been resolved.

Command: show system alarm history, Expected Output: Alarm history, including alarm codes, timestamps, and severity levels., Description: Verify that the ALARM_7159 code has been cleared and no new alarms are present.

Command: show hardware alarm, Expected Output: Hardware alarm details, including alarm codes, timestamps, and severity levels., Description: Verify that no hardware alarms are present to ensure the fault has been resolved.
```

Roll-back Procedures

- Step: Revert hardware changes, Command: hw-module revert, Description: Revert any hardware changes made during the resolution process to ensure the node returns to its original state.

- Step: Revert power supply redundancy settings, Command: power supply redundancy off, Description: Revert any power supply redundancy settings changed during the resolution process to ensure the node returns to its original state.